

Common fields, enums and types

Publishers, datasets, and venues

We use a few different terms to describe our data:

- A **dataset** is a source of data.
- A **venue** is an exchange, OTC market (e.g., ATS, ECN) or reporting entity.
- A **publisher** is a specific venue from a specific dataset.

See also

Read our [Venues and publishers](#) section for a more detailed explanation.

Publisher identifiers

All of our schemas include a `publisher_id` field, which is a unique numeric ID assigned by Databento to each publisher. A full list of publishers can be found using the [metadata.list_publishers](#) endpoint.

Venue and dataset identifiers

Each publisher is also assigned a string identifier (e.g., `OPRA.PILLAR.XCBO`), composed of two parts:

- **Dataset ID** (e.g., `OPRA.PILLAR`). This is used as the `dataset` argument in any API or client method. Dataset IDs can be found on the Databento portal on each dataset's [details page](#) or via the [metadata.list_datasets](#) endpoint.
- **Venue** (e.g., `XCBO`). For most markets, this is its [ISO 10383 MIC code](#) [↗](#), which is guaranteed to be four characters long. For entities without a MIC code, this string is arbitrarily assigned by Databento and will also be four characters long.

Instrument identifiers

All of our schemas contain an `instrument_id` field which is a numeric ID that maps to a given instrument. In most cases, this numeric ID is assigned by the publisher. For publishers that do not assign this value, we create a synthetic mapping for it.

`instrument_id` is only guaranteed to be unique within a given day. Some publishers provide a different instrument ID on different days for the same underlying instrument. Other publishers may use the same instrument ID for different underlying instruments at different points in time.

Depending on the use case, it may be easier to work with other symbology types such as `raw_symbol`. Our [symbology](#) documentation outlines the various symbology types we support.

Timestamps

All the timestamps in our data are expressed as the number of nanoseconds since the [UNIX epoch](#) [↗](#), i.e. UNIX timestamps. All timestamp fields are prefixed with `ts_`. Some of our timestamps are encoded as the difference, i.e. delta, relative to another timestamp. Such timestamp fields are suffixed with `_delta`.

We provide four types of timestamps, through the following fields:

- Event timestamps, `ts_event`
- Publisher sending timestamp, `ts_in_delta`
- Databento receive timestamp, `ts_recv`
- Databento sending timestamp (live only), `ts_out`

`UNDEF_TIMESTAMP` (`UINT64_MAX`, 18446744073709551615) is used to denote a null or undefined timestamp.

The event and publisher sending timestamps are provided by the publisher (or market), and we provide their original values without any adjustment.

ts_event

Most users will only need the event timestamp, i.e. `ts_event`. For market data, this represents the time that the event is received by the matching engine (tag 60 in FIX encoding).

The exact location where this timestamp is taken varies with matching engine architecture of each market. Some markets will handle different subsets of instruments on independent order gateways, while other markets will load balance the same subset of instruments across independent order gateways. Some markets take the event timestamp at the time it is received on the independent order gateways, while others may take this timestamp at the time it reaches a FIFO matching queue. In the former case, the clocks on independent order gateways are often not properly synchronized to the same clock source. Since we do not